Data Science, Statistics and the Future of PROs

40-minute roundtable

0–7 **INTRODUCTIONS**	*"We thought we'd keep this pretty informal. Maybe we can start by going around the table – your name, where you're from, your background, and what brought you to this discussion."*
7–12 **WHAT ARE PEOPLE DOING?**	• Is anyone already using data science or machine learning in PRO/QoL work? What are you using it for? • Anything you've seen recently that made you think it could be useful in your own work?
12–32 **DISCUSSION**	• **Opportunities:** Where does data science add something to what we already do? • **Training:** Good places to get started for learning DS/biostats? • **Software:** Differences between data science and statistics? • **New data:** What new sources are people working with or interested in – wearables, phones, text/voice, imaging, EHR, omics? • **Prediction:** Where is the greater opportunity – using PROs as predictors, or making patient experience the outcome we predict? • **Complement or challenge:** Where do these approaches fit comfortably with how we already do PRO research, and where do they challenge our usual assumptions or standards? • **Barriers and concerns:** What's making this difficult right now? And what worries you as these approaches become more common? • **Working together:** What do data scientists need to understand about PRO/QoL research? What does our field need to learn from data science?
Any time **IF WE NEED THEM**	• What are you excited about that you couldn't realistically do five years ago? • Where are you most skeptical? • What changes when the way we ask or score questions becomes adaptive? • What should validation look like when models are updated over time or used for individual patients? • When is machine learning not the right tool? • Are our current PRO datasets suitable for the models people want to build? • How do we keep the patient involved when more information comes from passive or derived data? • What would convince you that one of these tools was ready to use in practice? • What expertise or collaboration would help you do something you can't do now?
32–40 **WRAP-UP**	*"Before we finish, maybe one quick last round: what's one question you're leaving with, or one thing you'd like to try?"*

ROUNDTABLE FOCUS

How do established statistical and psychometric methods fit alongside emerging data science approaches? As patient-centered research incorporates data from wearables, smartphones, sensors, and other complex sources, where do these approaches complement existing PRO methods, and where do they challenge them?